

A Minimal Magnetar-Based Interpretation of GRB 250702B in a Disturbed Host Galaxy

Steve Girard

Independent Researcher, Québec, QC, Canada

Abstract

GRB 250702B displays an unprecedented combination of properties: three bright γ -ray episodes separated by a quasi-regular interval of $P \approx 2825$ s, a total duration of ~ 7 h, and an extremely obscured afterglow located ~ 1 kpc off-nucleus in a massive, dust-rich, morphologically disturbed host galaxy. Standard black-hole accretion models require finely tuned jet precession or multi-component engines to reproduce this behavior. We propose an alternative interpretation in which a single newborn magnetar, formed from the collapse of a stripped helium star, is embedded in a non-axisymmetric, merger-driven environment. Clumpy fallback accretion provides intermittent energy release, while slow precession of the jet or fallback disk—driven by torques from the asymmetric host potential—produces the observed ~ 47 -min recurrence through geometric modulation rather than intrinsic engine periodicity. The extreme extinction and off-nucleus location arise naturally from the local dust-rich environment. This minimal model accounts for the duration, quasi-periodicity, and obscuration of GRB 250702B without invoking a black hole or a highly fine-tuned jet structure.

Introduction

GRB 250702B exhibits temporal and environmental properties that place it outside the parameter space of standard long-duration gamma-ray bursts. Its multi-episode structure, hour-scale recurrence, and extreme obscuration challenge collapsar and black-hole accretion models, which typically require structured jets, intermittent feeding, or rapid precession to reproduce such behavior. The burst’s location in a strongly asymmetric, dust-rich, merger-driven host galaxy further suggests that environmental dynamics may play a central role in shaping the observed emission. Motivated by these considerations, we explore a minimal interpretation based on a single newborn magnetar embedded in a non-axisymmetric gravitational potential.

Results

This Letter focuses on the physical interpretation of GRB 250702B rather than on a full re-analysis of the observational data. We summarize the key empirical properties rel-

evant to the proposed model: (1) three bright γ -ray episodes separated by $P \approx 2825$ s, (2) a total duration of ~ 7 h, (3) an extremely obscured afterglow, and (4) an off-nuclear position in a disturbed, dust-rich host galaxy.

Discussion

The temporal structure of GRB 250702B—three bright γ -ray episodes separated by a quasi-regular interval of $P \approx 2825$ s and a total duration of ~ 7 h—poses a challenge to standard collapsar and black-hole accretion models. Reproducing such behavior within a BH-jet framework typically requires multiple finely tuned ingredients (multi-component jets, rapid precession, intermittent accretion, structured angular emission), none of which are independently required by the data. We therefore consider an alternative interpretation involving a single newborn magnetar embedded in a dynamically disturbed, dust-rich host galaxy.

In this picture, the collapse of a massive, stripped helium star forms a rapidly rotating, highly magnetized neutron star with a rotational energy reservoir of order 10^{52} erg. The progenitor lies ~ 1 kpc off-nucleus in a strongly asymmetric, merger-driven environment. Such regions naturally exhibit non-axisymmetric gravitational potentials, clumpy gas distributions, and substantial dust lanes. These conditions favor the development of a fallback structure that is neither smooth nor axisymmetric. Material returns to the compact object in discrete clumps, with characteristic viscous timescales of 10^3 – 10^4 s, consistent with the observed separation between γ -ray episodes.

A central feature of this scenario is that the ~ 47 -min recurrence does not require an intrinsic “engine clock.” Instead, the periodicity arises from slow precession of the jet or fallback disk. A misaligned disk subject to torques from the asymmetric host potential can precess on timescales comparable to those observed. If the jet direction is tied to the magnetic or disk axis, γ -ray emission becomes visible only when the precessing cone intersects our line of sight, producing quasi-periodic pulses even if the central engine remains continuously active. This geometric modulation removes the need for repeated engine shutdowns and restarts.

The extreme obscuration of the afterglow follows naturally from the off-nuclear location within a region of high dust column density, consistent with JWST imaging. In this con-

text, the rarity of GRB 250702B reflects the requirement that a magnetar-forming collapse occur within a highly perturbed, merger-driven environment capable of inducing both clumpy fallback and significant precession.

Although uncertainties remain—particularly regarding the stability of the precession over multiple cycles—this framework provides a parsimonious explanation for the long duration, quasi-periodic structure, extreme extinction, and off-nucleus position of GRB 250702B without invoking a black hole or a highly fine-tuned jet configuration. Further modeling of fallback dynamics in non-axisymmetric galactic potentials will be essential to assess the robustness of this scenario.

At the redshift of the host, any late-time energy injection from a newborn magnetar would be temporally stretched and shifted into the cm–mm bands, making radio facilities such as VLA and MeerKAT uniquely suited to detect a deviation from a standard synchrotron decay. A late-time flattening or rebrightening would strongly favor a magnetar engine over a black-hole accretion scenario.

Conclusions

A single newborn magnetar embedded in a disturbed, dust-rich host galaxy provides a minimal and physically motivated explanation for the unusual properties of GRB 250702B. Slow precession driven by the asymmetric gravitational potential offers a natural origin for the ~ 47 -min recurrence, while clumpy fallback accretion accounts for the multi-episode structure and long duration. This scenario avoids the fine-tuning required by black-hole models and highlights the importance of environmental dynamics in shaping extreme transients.